

Back Propagation:

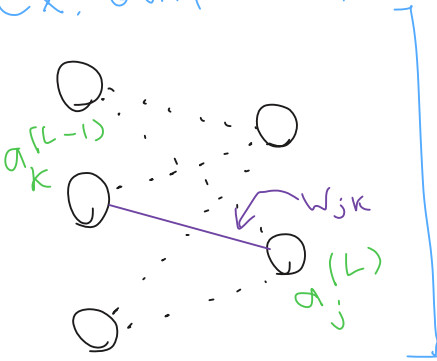
Goal:

Change parameters to minimize Cost Function via calculating gradient and moving in direction of steepest decrease.

Steps $\Rightarrow$

- Find $\frac{\partial C}{\partial w}$ and $\frac{\partial C}{\partial b}$
- Apply $w = w - \eta \frac{\partial C}{\partial w}$
and $b = b - \eta \frac{\partial C}{\partial b}$ for all w, b

ex. output layer



For a weight w_{jk} ,
we need to derivate
a formula for $\frac{\partial C}{\partial w_{jk}}$.

$$C = \sum_{j=0}^{n_{out}-1} (a_j - y_j)^2 \rightarrow \frac{\partial C}{\partial a_j} = \sum_{j=0}^{n_{out}-1} 2(a_j - y_j)$$

$$a_j = \sigma(z_j) \rightarrow \frac{\partial a_j}{\partial z_j} = \sigma'(z_j)$$

$$z_j = \left(\sum_{k=0}^{n-1} a_k^{prev} \cdot w_{jk} \right) + b_j \rightarrow \frac{\partial z_j}{\partial w_{jk}} = a_k$$

Chain Rule:

$$\frac{\partial C}{\partial w_{jk}} = \frac{\partial C}{\partial a_j} \cdot \frac{\partial a_j}{\partial z_j} \cdot \frac{\partial z_j}{\partial w_{jk}}$$